PYTHON - Quick Handwritten Notes

What is Python?

Python is a high-level, interpreted and general-purpose programming language. It is easy to read and beginner friendly.

Main features

-> Simple syntax -> Interpreted -> Portable
-> Open source -> Large libraries -> Object-oriented support

Basic building blocks

Variable = a name that stores/refers to a value.
Data types: int, float, str, bool, list, tuple, dict, set.
input() takes user input. print() displays output.

```
name = input("Enter name: ")
age = int(input("Enter age: "))
print("Hello", name)
print("Age =", age)
```

Operators

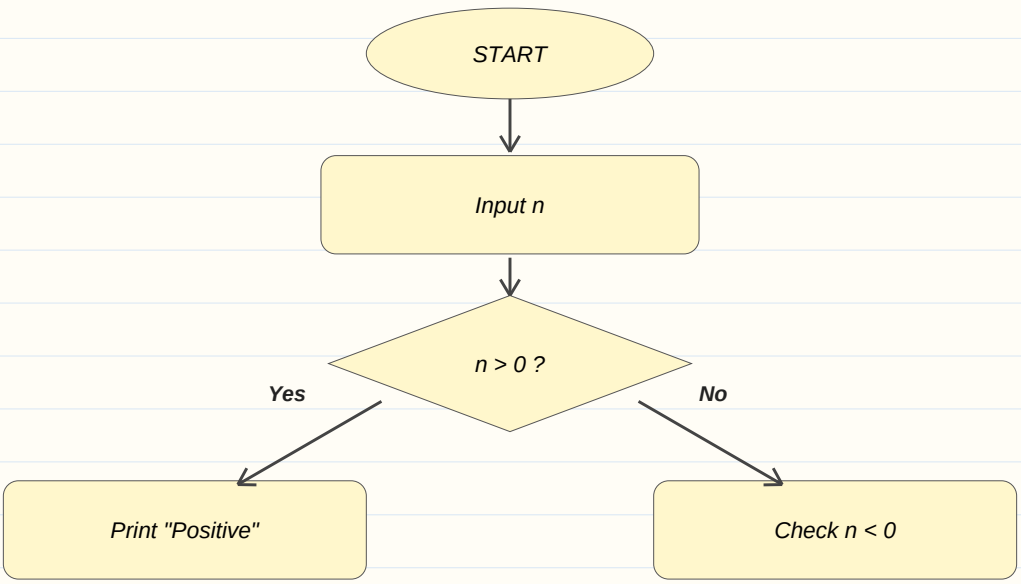
*Arithmetic: + - * / // % **
Comparison: == != > < >= <=
Logical: and or not*

CONDITIONS + Flow Diagram

Decision making

if checks a condition. elif checks another condition.
else runs when previous conditions are False.

```
n = int(input("Enter number: "))
if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```



FOR LOOP - Explanation + Programs

for loop

Used to repeat code over a sequence or range.
range(start, stop, step) - stop value is NOT included.

```
for i in range(1, 6):
    print(i)

# Output: 1 2 3 4 5
```

Program: Multiplication table

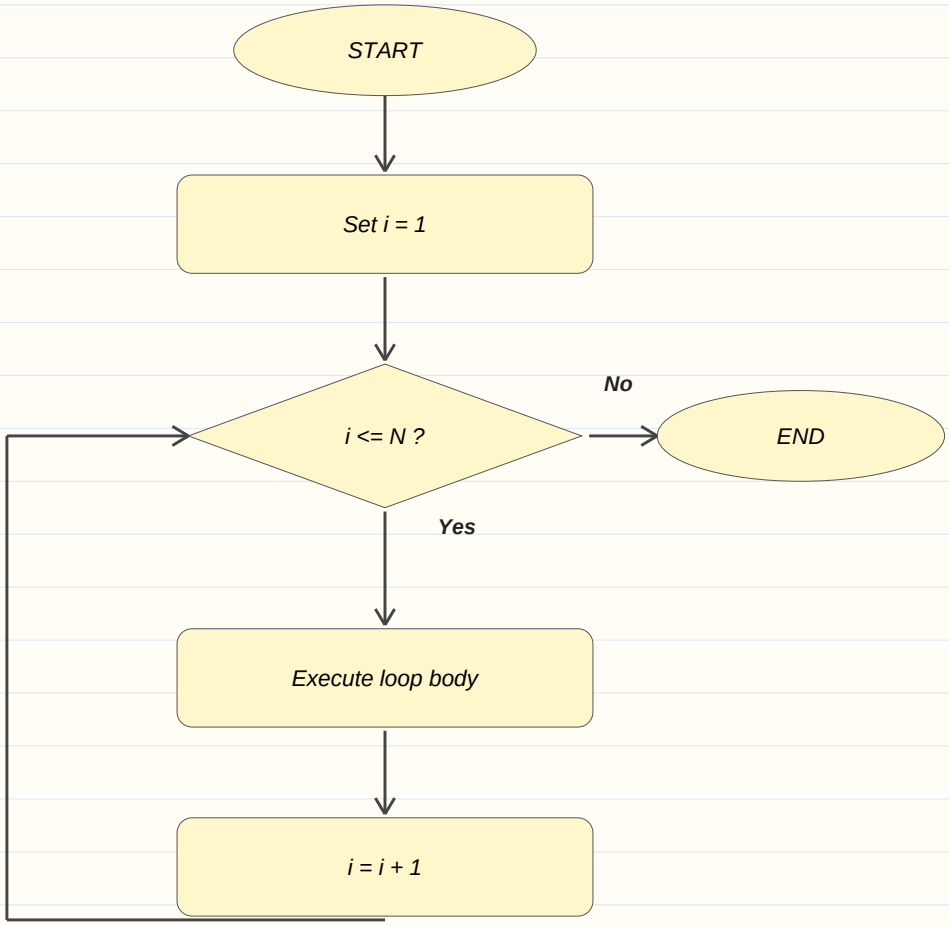
```
n = int(input("Enter number: "))
for i in range(1, 11):
    print(n, "x", i, "=", n*i)
```

Program: Sum 1 to N

```
n = int(input("Enter N: "))
total = 0
for i in range(1, n+1):
    total += i
print("Sum =", total)
```

Remember: indentation tells Python which statements belong to the loop.

FOR LOOP - Flowchart



Trace example: range(1, 4)

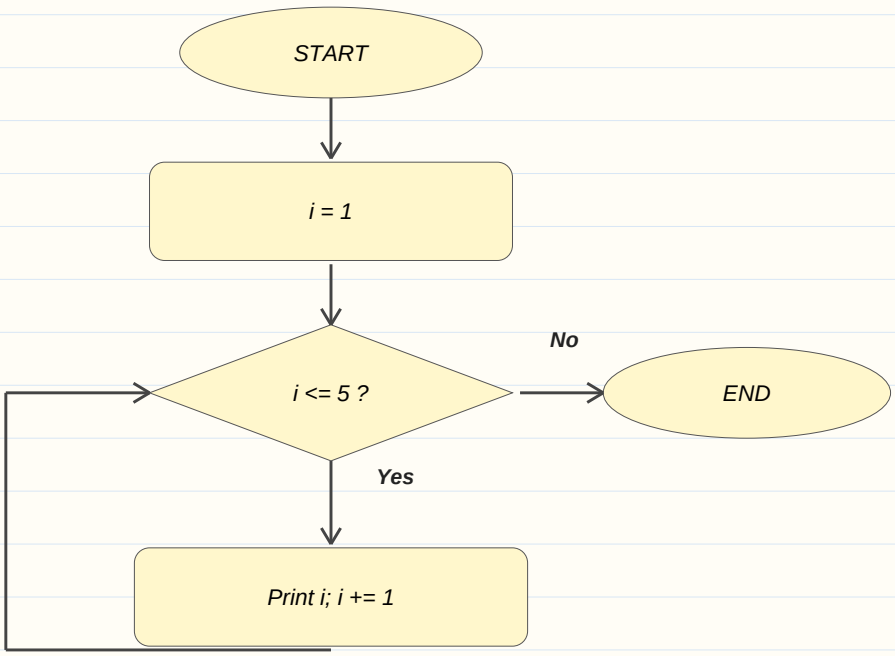
i = 1 -> body runs
i = 2 -> body runs
i = 3 -> body runs
i = 4 -> condition/range ends

WHILE LOOP - Explanation + Flowchart

while loop

Repeats while a condition is True. Update the loop variable, or the loop may never stop (infinite loop).

```
i = 1
while i <= 5:
    print(i)
    i += 1
```



IMPORTANT PROGRAMS

1) Even or Odd

```
n = int(input("Number: "))
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

2) Factorial

```
n = int(input("Number: "))
fact = 1
for i in range(1, n+1):
    fact *= i
print(fact)
```

3) Prime Number

```
n = int(input("Number: "))
prime = n >= 2
for i in range(2, n):
    if n % i == 0:
        prime = False
        break
print("Prime" if prime else "Not prime")
```

4) Factors

```
n = int(input("Number: "))
for i in range(1, n+1):
    if n % i == 0:
        print(i)
```

LOOP CONTROL + EXAM CHEAT SHEET

break

Stops the nearest loop immediately.

continue

Skips the remaining code in the current iteration.

pass

Placeholder statement - it performs no action.

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)   # 1, 2, 4, 5
```

Common mistakes

1. *Missing ':' after if/for/while.*
2. *Wrong indentation.*
3. *Using = instead of == for comparison.*
4. *Forgetting int(input(...)) for numbers.*
5. *Forgetting to update a while-loop variable.*
6. *Expecting range stop value to be included.*